SustainX Innovation Challenge 2026

Workshop Session: What You Will Build in SustainX 2026

From **Energy Data** → **Digital Value** → **User Transactions**

Where Innovation Meets Infrastructure

SUSTAIN X
2026

Why This Session Is Critical for Your Hackathon

- Clearly understand what you are expected to build
- Avoid wasting time on the wrong ideas
- Start the hackathon with a strong, focused direction

Imagine This

- You generate solar energy at home
- Your excess energy is converted into digital value
- You go to a supermarket or take the metro
- You pay using that energy

Why This Matters For Mauritius

- Increasing adoption of solar energy
- Limited ways to use excess energy
- Opportunity to create a new digital energy economy

What This Challenge Is About

- Designing systems where energy becomes usable value
- Creating solutions for real-world energy usage
- Building practical, user-focused applications

Why This Is Happening

Energy systems are changing:

- Energy is becoming decentralized
- People are becoming energy producers

The REAL Challenge

Designing a system that:

- Converts energy into digital value
- Allows users to store and use that value
- Enables real-world transactions

What You Are Building

A system that includes:

- Energy data → converted into digital value
- A user app to view balance and make transactions
- A backend/system that records and updates everything

How Your System Should Work

Your system should follow this flow:

- Energy data (input)
- Conversion into value
- User wallet (display)
- Transactions (usage)

Understanding the Dataset

Dataset includes:

- User ID – identifies the user
- User Type – producer or consumer
- Import (kWh) – energy used
- Export (kWh) – energy produced

How You Will Use This Data

You will use this data to:

- Identify how much energy each user generates
- Calculate the digital value created
- Assign value to each user (balance)
- Update balances after transactions

Digital Units (Core Concept)

Your system must:

- Convert energy into digital value
- Represent that value as units (e.g., 1 KWH = 1 Energy Credit = 1 EC)
- Store it in a user wallet
- Display user balances
- Allow payments or transfers

Example Flow

- User generates energy
- System converts it into digital value
- Value is added to user wallet
- User can now spend or transfer it

Transactions

Your system must allow:

- Sending value from one user to another
- Deducting value from sender
- Adding value to receiver
- Updating balances after each transaction

System Integrity

Ensure your system:

- Does not allow negative balances
- Updates balances correctly after each transaction
- Prevents invalid actions (e.g., spending more than available)

Traceability

Your system should show:

- A record of all transactions
- Visible balance updates
- Clear logs of system actions

System Architecture (Big Picture)

Your system should include:

- Energy data as input
- A logic to convert it into value
- A system to store and update balances
- A user interface (wallet / app)
- A way to perform transactions

What a Winning System Looks Like

A strong solution will:

- Clearly convert energy data into digital value
- Show how value is calculated (e.g., export – import)
- Display user balances in a simple wallet
- Allow transactions between users
- Work consistently with different datasets

Weak vs Strong Solution

❌ Weak Solution

- Only displays raw data
- No clear logic or value conversion
- No digital value (credits) generated
- No transaction capability
- Difficult to explain

✅ Strong Solution

- Converts data into digital value (e.g., export – import)
- Clear and logical system flow
- Tracks user balances (wallet)
- Supports transactions between users
- Easy to explain and demonstrate

What Judges Look For

Judges will look for:

- Clear explanation of your system
- Logical flow (input → processing → output)
- Correct use of the dataset
- Clear value conversion (e.g., export – import)
- A working and practical solution

Common Mistakes

Avoid:

- Not using the dataset to drive your logic
- No clear value conversion (data → digital value)
- No clear system flow
- Overcomplicating the solution
- Weak or unclear explanation

Winning Strategy

To stand out:

- Build a clear and logical system (input → processing → output)
- Show how data becomes value (export – import → credits)
- Use the dataset correctly and consistently
- Keep your solution simple and easy to understand
- Clearly explain and demonstrate your system

Think Real-World

Ask yourself:

- Would this system work with real energy data?
- Can it handle new datasets (e.g., monthly updates)?
- Are the transactions logical and consistent?
- Can I clearly explain how value is created and used?

Final Message

Think clearly
Build logically
Turn data into value
Explain simply

Thank You

Answer Time

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